

A Detoxified Extract of *Rhus verniciflua* Stokes Upregulated the Expression of BDNF and GDNF in the Rat Brain and the Human Dopaminergic Cell Line SH-SY5Y

Kumar SAPKOTA,^{1,5,*} Seung KIM,^{2,*} Myung-Kon KIM,³ and Sung-Jun KIM^{1,4,†}

¹Department of Biotechnology, Chosun University, Gwangju, 501-759, Republic of Korea

²Department of Alternative Medicine, Gwangju University, Gwangju, 503-703, Republic of Korea

³Department of Bio-Food Technology, Chonbuk National University, Iksan, 570-752, Republic of Korea

⁴BK21 Research Team for Protein Activity Control, Chosun University, Gwangju, 501-759, Republic of Korea

⁵Central Department of Zoology, Tribhuvan University, Kirtipur, Kathmandu, Nepal

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Rhus verniciflua Stokes (RVS) has traditionally been used as a food supplement and a traditional herbal medicine for centuries in Korea. This study attempted to evaluate the effects of RVS on the expression of Brain derived neurotrophic factor (BDNF) and Glial cell line-derived neurotrophic factor (GDNF) in SH-SY5Y cells and the rat brain. The results indicated that RVS is a potent inducer of Neurotrophic factor (NTF) production both *in vitro* and *in vivo*. Treatment with 10 μ g/ml and 10 mg/kg RVS for 4 h of SH-SY5Y cells and rats yielded significant increases in BDNF and GDNF protein levels. We also detected BDNF and GDNF immunoreactive neurons in the rat brain. Both BDNF and GDNF-immunohistochemical staining was markedly enhanced in the animals treated with RVS. These results suggest that RVS serves as an ideal adjuvant in regard to regulating NTF expression, and can contribute to neuroprotection in neurodegenerative diseases.

Key words: gene expression; neurotrophic factors; neuroprotection; Parkinson's disease; *Rhus verniciflua*

Neurotrophic factors (NTFs) are secreted proteins that have attracted increasing attention because, in addition to promoting the survival and differentiation of developing neurons, they appear to be necessary for the synaptic activity and plasticity of mature neurons.^{1,2} Brain-derived neurotrophic factor (BDNF) and glial cell line-derived neurotrophic factor (GDNF) belong to two different families of NTFs. They regulate diverse neuronal functions and have been identified as critical to the survival of dopaminergic (DA) neurons.³ BDNF is a secretory protein. It mediates its effects *via* two classes of receptors: a member of the tropomyosin receptor kinase (trk) family of receptors, trkB, and a member of the tumor necrosis factor family of receptors, the p75 neurotrophin receptor.^{4,5} GDNF belongs to a subfamily within the transforming growth factor-superfamily, together with neurturin, persephin, and artemin. In contrast to BDNF, which binds to two distinct classes

of receptors, GDNF mediates its effects *via* a single receptor complex composed of a ligand-binding glycosylphosphatidyl inositol membrane-linked receptor subunit, named GDNF family receptor α (GDNFR α), and a signal-transducing tyrosine kinase subunit, Ret.⁶ It has been suggested that changes in the levels or distribution of these NTFs are important in the pathogenesis of neurodegenerative conditions in humans.⁷ Moreover, it has been found that both BDNF and GDNF can protect nigrostriatal DA neurons from neurotoxins in animal models of Parkinson's disease (PD).^{8–11} PD is the most common neurodegenerative movement disorder, affecting more than six million people worldwide,¹² and is expected to impose an ever increasing impact on society emotionally, socially, and financially.

Recently, natural bioactive substances have received a great deal of attention in the treatment of neurodegenerative disorders. Examination of ethnopharmacologic treatments might be a cost-effective alternative, since naturally occurring compounds have been found to be beneficial in PD.^{13–15}

Rhus verniciflua Stokes (RVS) of the Anacardiaceae family is commonly known as the lacquer tree. It is used as a food additive and traditional herbal medicine for cancer and stomach diseases.¹⁶ RVS contains phenolic compounds, including gallic acid, protocatechuic acid, quercetin, fustin, fisetin, sulfuretin and butein, and has been reported to exert antioxidant,^{17,18} anti-inflammatory,¹⁹ antitumorigenic,²⁰ anti-mutagenic,²¹ and anti-apoptotic²² effects. However, its application as a food additive or source of pharmaceutical compounds has been limited because urushiol congeners in the plant cause irritation, inflammation, or blistering in sensitive individuals, a reaction known as urushiol-induced contact dermatitis.²³ Hence it is important to remove urushiol from the substrate before using it as a food additive or source of pharmaceutical compounds. We have previously developed a new biological approach to the removal of urushiol from RVS using mushrooms,²⁴ and prepared detoxified RVS and evaluated its effect on the expression of catecholamine biosynthetic enzymes.²⁵

[†] To whom correspondence should be addressed. Department of Biotechnology, Chosun University, 375 Seosuk-dong, Dong-gu, Gwangju, 501-759, Republic of Korea; Tel/Fax: +82-62-230-6664; E-mail: sjbkim@chosun.ac.kr

* These authors contributed equally to this study.

In this study, we examined the effects of this detoxified RVS on the expression of BDNF and GDNF in the rat brain and a cultured human dopaminergic cell line SH-SY5Y.

Materials and Methods

Materials. Moloney-murine leukemia virus ribonuclease (M-MLV), oligo deoxythymidine (oligo-dT), deoxynucleic acid triphosphate (dNTPs), Taq polymerase, specific primers (for BDNF, GDNF, Actin, and GAPDH), and a 100 bp DNA ladder were purchased from Bioneer (Daejeon, Korea). Anti-BDNF, rabbit polyclonal immunoglobulin G (IgG), Anti-GDNF, mouse monoclonal, and secondary antibody-horse-radish peroxidase (HRP) conjugated goat-anti-mouse IgG, HRP conjugated goat-anti-rabbit IgG were from Santa Cruz Biotechnology (Santa Cruz, CA). Anti-actin antibody was from Biomedica (Foster City, CA). Normal goat serum (NGS), 3-3' diaminobenzidine (DAB), and Vectastain ABC reagents were from Vector (Burlingame, CA). The other reagents were of special grade, and were purchased commercially.

Plant material and preparation of RVS extract. Bark of *Rhus verniciflua* Stokes (Anacardiaceae) was purchased from a commercial supplier in Wonju, South Korea, and was authenticated by Professor Myung-Kon Kim, Department of Bio-Food technology, Chonbuk National University, South Korea. A voucher specimen was deposited at Chonbuk National University. One thousand g of sun-dried bark of RVS was crushed. The crushed material was then passed through a 10-mesh size sieve, and the allergenic urushiol from the extract was removed by previously described procedures.^{24,26-28} The yield was 82.6%. This extract was then dried in an oven at 65 °C for 3 d and ground and homogenized thoroughly. Two hundred g of dark-brown powder was then boiled with 2 liters of water for 10 h at 100 °C. The extract was filtered with Whatman paper (Whatman, Kent, UK), and the water removed by vacuum rotary evaporation. The concentrate was freeze-dried. The yield was 5.4%.

Cell culture and treatments. Human DA neuronal cell line SH-SY5Y was obtained from the American Tissue Culture Collection (ATCC, Rockville, MD). The cells were cultured in Dulbecco's Modified Eagle Medium: Nutrient Mixture F-12 (DMEM/F12, Gibco, Gaithersburg, MD) supplemented with 10% Fetal Bovine Serum (FBS) and penicillin (100 U/ml)-streptomycin (100 µg/ml) at 37 °C in 5% CO₂. The media were changed every 2 d. The cells were treated with saline or 10 µg/ml of RVS for 2, 4, 8, and 12 h respectively. The treated cells were harvested by trypsinization. In each experiment each treatment was performed in triplicate.

Animals and treatments. Male Sprague-Dawley rats (280–320 g) approximately 3 months old were used in all experiments. The animals were housed four per cage and were maintained under control environmental conditions (22 ± 2 °C, 12 h light/dark cycle). Food (Certified Rodent Diet 5002, Orient Bio, Seongnam, Korea) and tap water were available *ad libitum*. All efforts were made to minimize animal suffering and to reduce the number of animals used. All experimental procedures were performed in accordance with the NIH Guide for the Care and Use of Laboratory Animals (NIH publication no. 80-23 revised 1996) and related ethical regulations of Chosun University.

Before treatment, the rats were weighed, and RVS was prepared in phosphate buffered saline (PBS) to achieve the desired dose, which was based on the average weight of the animals. For the dose-response study, three rats/group were gavaged with 10, 20, and 40 mg/kg body weight RVS, and were killed 2, 4, 8, and 12 h after dosing. The control groups received equal volumes of PBS. For the time course study, the animals were treated by gavage with 1 ml of PBS for a nominal dose of 0 (vehicle control) or 10, 20, and 40 mg/kg of body weight of RVS. Three animals were treated per dose group, and rats were killed 2, 4, 8, or 12 h after dosing. An untreated group of rats was also included. They were killed at time 0 when the other animals were dosed. This dose and the time-course were chosen based on our preliminary experiments. The control and unstressed rats were sacrificed immediately after the removal from their home cages. They were anesthetized with mixture

Table 1. Primers Used for RT-PCR

Genes	Forward primer Reverse primer	PCR product size (bp)
BDNF (rat)	5'-AACGAAGAAAACCATAAGGA-3' 5'-GTCTATCCTTATGAATCGCC-3'	465
GDNF (rat)	5'-CAGAGAATTCCAGAGGGAAA-3' 5'-TACATCCACACCGTTTAGCG-3'	334
BDNF	5'-GATGACCATTCTTTTCCTTACTATGG-3' 5'-CTATCTTCCCTTTTAAATGGTCAAT-3'	745
GDNF	5'-CTGGGCTATGAAACCAAGGA-3' 5'-GACAGGTCATCATCAAAGGC-3'	173
GAPDH	5'-CCATGTTCGTATGGGTGTGAACCA-3' 5'-GCCAGTAGAGGCAGGGATGATGTTC-3'	251
β-Actin	5'-CCTCTATGCCAACACAGT-3' 5'-AGCCACCAATCCACACAG-3'	155

of Zoletil 50[®] (tiletamine plus zolazepam) (Virbac, Carros, France) and Rompun[®] (xylazine) (Bayer, Seoul, Korea) in solution (3:1 ratio, 1 ml/kg, i.p.) before sacrifice. The brains were rapidly excised, cleaned of fat tissue, frozen in liquid nitrogen, and stored at –70 °C for the further analysis.

RNA isolation and reverse transcription (RT). Total RNA was extracted from whole frozen brains of rats and cultured using TRIzol 'reagent (Invitrogen, Life Technologies, Carlsbad, CA) following the protocol provided by the company. Two µg of total RNA samples were reverse transcribed for each sample to be analyzed by incubation with a reverse transcription mixture containing the following constituents: 10 pmol oligo-dT primer, 1× PCR buffer, 0.1 M DTT, 10 mM dNTPs, 20 units of RNase inhibitor, and 200 units of M-MLV. The reaction mixture was incubated for 60 min at 42 °C, followed by 5 min at 70 °C to inactivate the RT enzyme. The quality of the cDNA was verified by PCR amplification of β-actin and by GAPDH.

Polymerase chain reaction (PCR) and analysis of PCR products. The cDNA in the RT product was amplified using Taq DNA polymerase. A PCR reaction was performed in 20 µl of the total volume using 10 pmol of the corresponding primers (Table 1). The cDNA was amplified under the following reaction conditions: denaturation at 94 °C for 30 s, annealing at 56 °C for 45 s for BDNF (rat), and 55 °C for BDNF (human), at 62 °C for GDNF (rat), and at 54 °C for GDNF (human), at 56 °C for β-actin, and at 58 °C for GAPDH, and polymerization at 72 °C for 30 s. The cyclic process was performed 35 times for BDNF, and GDNF, and 25 times for β-actin and GAPDH. The PCR products were analyzed on 1.2% agarose gel and visualized with ethidium bromide. The stained intensity of the individual bands was evaluated with Gel Quant software (DNR Bio-Imaging Systems, Jerusalem, Israel).

Western blot analysis. Protein was isolated with Trizol following the manufacturer's protocol. Subsequently, the protein concentration was determined using a bicinchoninic acid (BCA) protein assay kit (Pierce, Rockford, IL). Twenty µg of the protein extracts were separated by SDS-polyacrylamide gel electrophoresis and then transferred to a polyvinylidene difluoride (PVDF) membrane. After the transfer was complete, the membrane was blocked overnight with 3% bovine serum albumin (BSA) in Tris-buffered saline with Tween 20 (TBS-T) (10 mM Tris-HCl, 150 mM NaCl, and 0.1% Tween 20, pH 7.5) at room temperature to avoid non-specific binding of the antibody. Incubation with both primary and secondary antibodies was performed in TBS-T solution for 2 h at room temperature. The membrane was washed 3 × 10 min with TBS-T between primary and secondary antibody application and 3 × 15 min with TBS-T after incubation with the secondary antibody. The following primary antibody dilutions were used: anti-BDNF, 1:1,000, and anti-GDNF, 1:1,000. All secondary antibodies coupled to HRP were used at 1:400 dilution. The same amount of protein was loaded and reacted with anti-mouse monoclonal actin antibody (1:400), which was used as a control for protein loading. To reveal the reaction bands, the membrane was

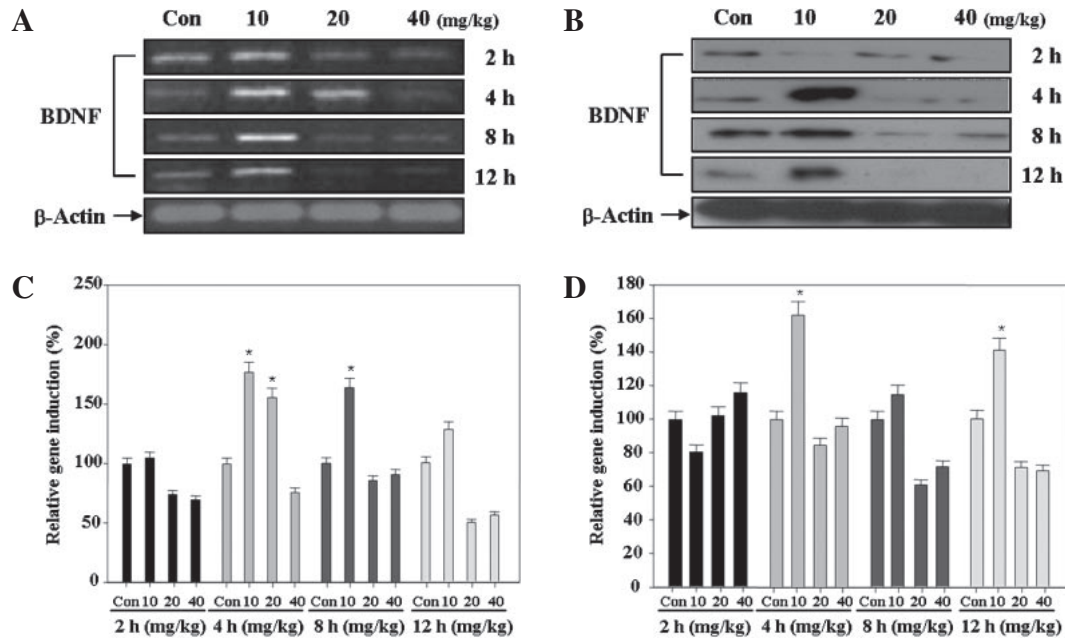


Fig. 1. Effects of Various Doses of RVS on BDNF mRNA and Protein Levels in the Rat Brain at Various Time Points.

Rats were treated with 10, 20, or 40 mg/kg RVS in saline for 2, 4, 8, or 12 h, as described in "Materials and Methods." BDNF mRNA and protein from brain tissue isolated from the animals were analyzed separately by RT-PCR and Western blot. Representative RT-PCR (A), Western blot (B) (upper panel), and quantification of relative intensity data (C, D) (lower panel) are shown for the BDNF mRNA and protein levels. The values were normalized to β -actin and are expressed as mean $\pm$ SD ($n = 3$), with the levels of BDNF mRNA and protein in the saline-treated control group taken as 100%. C, untreated control. * $p < 0.05$ versus the control group.

reacted with the WEST-ZOL (plus) Western blot detection system (Intron Biotechnology, Daejeon, Korea) and exposed on X-ray film (BioMax MS-1, Eastman Kodak, Rochester, NY). A digital image system was used to determine the density of the bands (Gel Quant, DNR Bio-Imaging Systems).

Immunohistochemistry. The rats were treated by gavage with 10 mg/kg of RVS. The control groups were treated with equal volumes of PBS. After 4 h, the rats of each group were deeply anesthetized with sodium pentobarbital (50 mg/kg, i.p.), and perfused transcardially with 0.1 M PBS (pH 7.4) followed by 4% paraformaldehyde in PBS for fixation of tissue. The brains were removed, and post-fixed in the same solution for 24 h at 4°C, followed by immersion in 10%, 20%, or 30% sucrose solution in PBS at 4°C until they sank. The whole brain was then cut into 35 μ m coronal sections using freezing cryostat (Leica, Nussloch, Germany) and collected in PBS. Endogenous peroxidase activity was inhibited by incubating the sections in 0.5% H_2O_2 in methanol. Non-specific binding sites were blocked by incubating the sections in PBS containing 1.5% normal goat serum (NGS), 0.5% BSA, and 0.1% Triton X-100. These sections were then incubated for 24 h in primary antibody (anti-BDNF and anti-GDNF antibody, 1:500). After removal of the primary antibody, the sections were washed 3 times with PBS and then incubated in peroxidase linked secondary antibody (1:200) for 2 h at room temperature, followed by three washes with PBS. Negative controls omitting the primary antibody were run in parallel. Color for the peroxidase linked antibody was developed with DAB as chromogen. The sections were transferred to gelatinized glass slides, dehydrated, cleared, mounted in DPX, and cover-slipped. Finally they were viewed under an inverted fluorescence microscope (Nikon TE2000, Tokyo), and photographs were taken.

Statistical analysis. The data were expressed as means $\pm$ SD. Statistical evaluation of the data was performed by Student's t -test. $p < 0.05$ was considered statistically significant.

Results

Induction of BDNF mRNA and protein

Figure 1 shows the dose-response and time-course effects of RVS on BDNF mRNA and protein level

expression. As shown in Fig. 1A and C, RVS at a dose of 10 mg/kg induced the expression of BDNF mRNA, and maximum levels of induction (1.7 and 1.6-fold) were observed after 4 and 8 h of treatment. After 12 h, a small induction was observed (about 1.2-fold). In contrast to the changes observed with low doses, high-dose (20 and 40 mg/kg) treatments were mostly ineffective. Significant changes were observed only with 20 mg/kg of RVS after 4 h of treatment, which raised the BDNF mRNA levels in the brain by about 1.5-fold. A high dose of RVS, 40 mg/kg, did not induce any additional changes in mRNA levels at any time point (Fig. 1A and C). After 12 h of treatment, both concentrations of RVS (20 and 40 mg/kg) reduced the BDNF mRNA levels, by approximately 49% and 46% respectively.

To estimate the protein level of BDNF, Western blot analysis was carried out. As indicated in Fig. 1B and D, the most effective concentration was 10 mg/kg, which yielded significant increases in BDNF protein levels, of about 1.6-fold above the saline-treated controls, after 4 h of treatment. Treatments with the same dose of RVS led to an approximately 1.1 to 1.4-fold elevation in BDNF protein levels in the brain after 8 and 12 h respectively. Higher concentrations (20 and 40 mg/kg) of RVS were ineffective, and did not produce any differential expression of protein. These results are consistent with the results of RNA analysis.

Next we evaluated the *in vitro* induction of BDNF mRNA and protein due to RVS in SH-SY5Y cells. As shown in Fig. 2A and C, slight induction of the BDNF mRNA level of cells was observed 2 h after RVS treatment, with maximum expression (about 1.07-fold) after treatment for 4 h. After 8 and 12 h, very weak expression was seen. Similarly to our findings for mRNA levels, 4 h of treatment with RVS at 10 μ g/ml induced BDNF protein significantly, by 2.2-fold, but

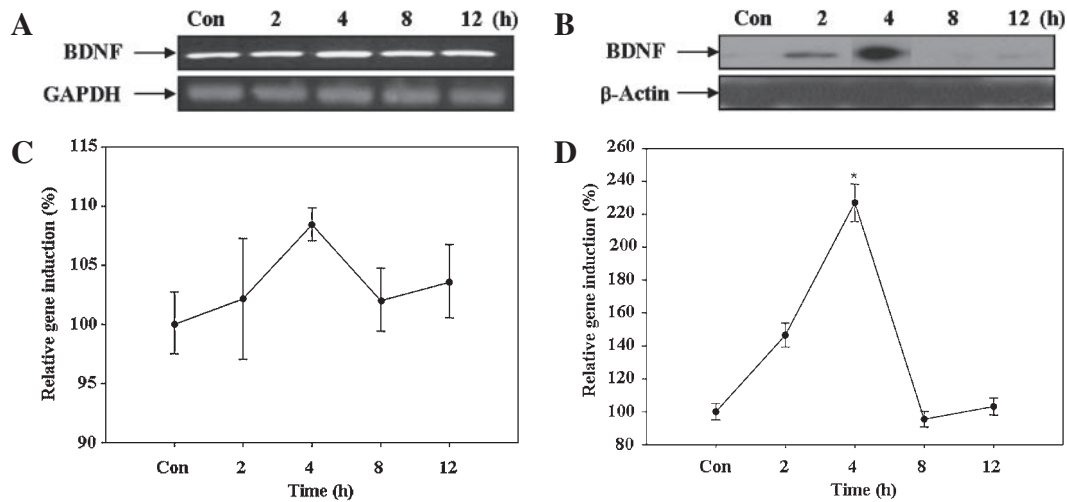


Fig. 2. Effects of RVS on the mRNA and Protein Levels of BDNF in SH-SY5Y Cells.

Cells were treated with 10 µg/ml of RVS for 2, 4, 8, or 12 h, as described in "Materials and Methods." Each treatment group was analyzed separately by RT-PCR and Western blot. Representative RT-PCR (A), Western blot (B) (upper panel), and quantification of relative intensity data (C, D) (lower panel) are shown for the BDNF mRNA and protein levels. The values are normalized to GAPDH and β-actin and as expressed as mean ± SD (n = 3) with levels of BDNF mRNA and protein in the saline-treated control group taken as 100%. C, untreated control. **p* < 0.05 versus the control group.

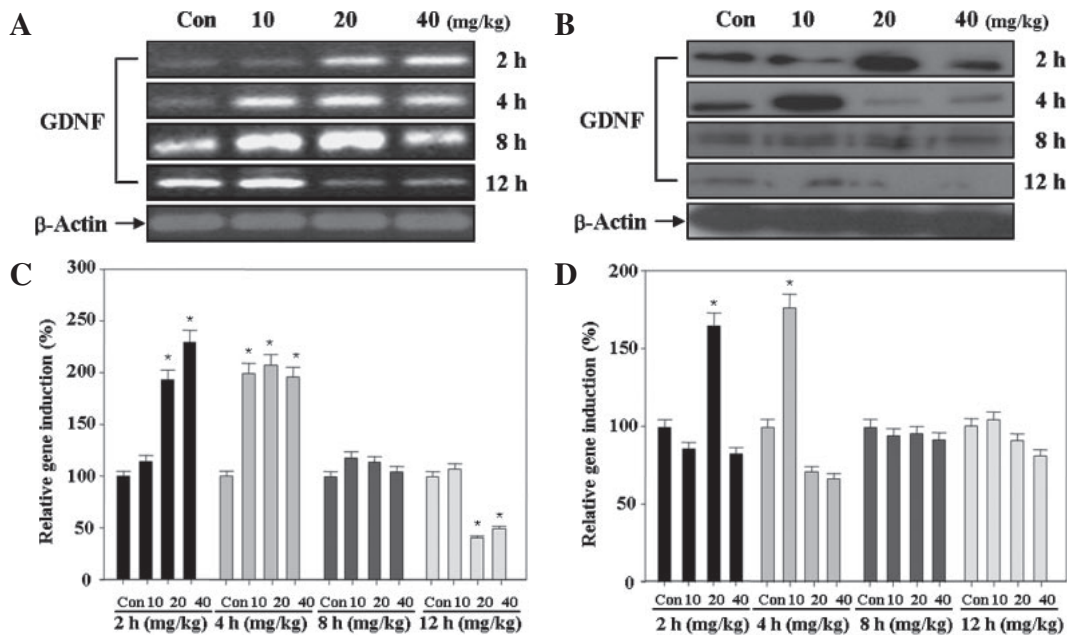


Fig. 3. Effects of Various Doses of RVS on the GDNF mRNA and Protein Levels in the Rat Brain at Various Time Points.

Rats were treated with 10, 20, or 40 mg/kg RVS in saline for 2, 4, 8, or 12 h, as described in "Materials and Methods." GDNF mRNA and protein from the brain tissue isolated from the animals were analyzed separately by RT-PCR and Western blot. Representative RT-PCR (A), Western blot (B) (upper panel), and quantification of relative intensity data (C, D) (lower panel) are shown for the GDNF mRNA and protein levels. The values are normalized to β-actin and are expressed as mean ± SD (n = 3) with the levels of GDNF mRNA and protein in the saline-treated control group taken as 100%. C, untreated control. **p* < 0.05 versus the control group.

this effect was no longer found at 8 or 12 h (Fig. 2B and D).

Induction of GDNF mRNA and protein

The effects of various concentrations of RVS at different time points on GDNF expression were investigated at both the mRNA and the protein level. Figure 3A and C illustrates the expression pattern of GDNF mRNA in response to various concentrations of RVS at different time points. We found that treatment with RVS at 10 mg/kg resulted in GDNF mRNA expression changes in the rat brain at all time points. The maximum levels of the GDNF mRNA were about

2-fold of the control at 4 h after treatment, and after 12 h, they returned to the basal level almost completely. RVS at a dose of 20 mg/kg led to an approximately 1.9- to 2-fold elevation of GDNF mRNA levels after treatment for 2 and 4 h. After 8 h no differential expression was seen. A similar pattern of expression was also observed for RVS at 40 mg/kg, with maximum induction (2.2-1.9-fold) at 2 and 4 h. However, 12 h after treatment (20 and 40 mg/kg), the levels of GDNF mRNAs were significantly down (*p* < 0.05). In contrast to the changes observed for short-term treatment, long-term treatments were mostly ineffective at high doses of RVS.

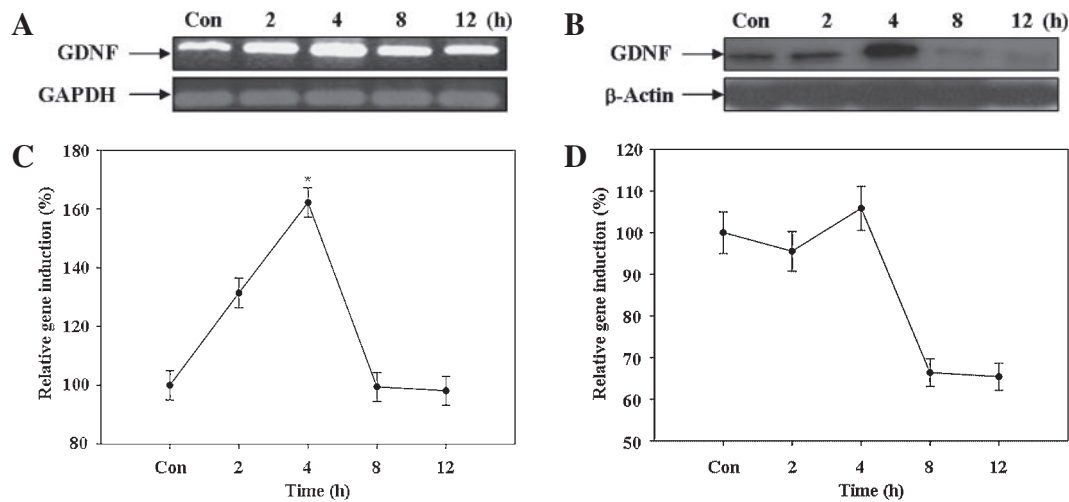


Fig. 4. Effects of RVS on the mRNA and Protein Levels of GDNF in SH-SY5Y Cells.

Cells were treated with 10 μ g/ml of RVS for 2, 4, 8, or 12 h, as described in "Materials and Methods." Each treatment group was analyzed separately by RT-PCR and Western blot. Representative RT-PCR (A) Western blot (B) (upper panel), and quantification of relative intensity data (C, D) (lower panel) are shown for the GDNF mRNA and protein levels. The values are normalized to GAPDH and β -actin and are expressed as mean $\pm$ SD ($n = 3$) with the levels of GDNF mRNA and protein in the saline-treated control group taken as 100%. C, untreated control. * $p < 0.05$ versus the control group.

Western blot analysis (Fig. 3B and D) indicated that induction of GDNF protein in the brain was more effective at 10 and 20 mg/kg of RVS than at 40 mg/kg. Both concentrations of RVS significantly elevated proteins for this gene, but at different time points. For example, treatment with 10 mg/kg raised the GDNF protein levels by about 1.7-fold at 4 h, while a similar induction pattern was found for 20 mg/kg RVS after 2 h of treatment. In contrast to this strong upregulation of the GDNF protein after treatment at these doses, this protein was not induced by a high RVS concentration (40 mg/kg).

Figure 4 shows the effects of RVS on GDNF mRNA and protein level expression in SH-SY5Y cells. In the control group, we found a low level of GDNF mRNA expression. Treatment with RVS for 2 h induced visible upregulation of GDNF mRNA expression. A typical bell-shaped time-course curve was observed, the most effective time point being at 4 h, during which average expression was 1.6-fold higher than control. The effect of upregulation extinguished 8 h after RVS treatment. A similar pattern of expression was also observed for protein levels, although the level of expression was lower than for mRNA.

Immunohistochemical analysis of BDNF and GDNF gene expression due to RVS

Expression of the BDNF and GDNF enzymes in the rat brain was evaluated by immunohistochemistry. The results indicated that BDNF and GDNF immunoreactive neurons are located in the substantia nigra of the rat brain (Fig. 5). BDNF and GDNF immunohistochemical staining was remarkably different in the RVS-treated groups as compared to the saline-treated animals.

Discussion

RVS is a well-known medicinal plant in Korea. It has multiple therapeutic functions, including antioxidant activity, but much less is known about its effects on

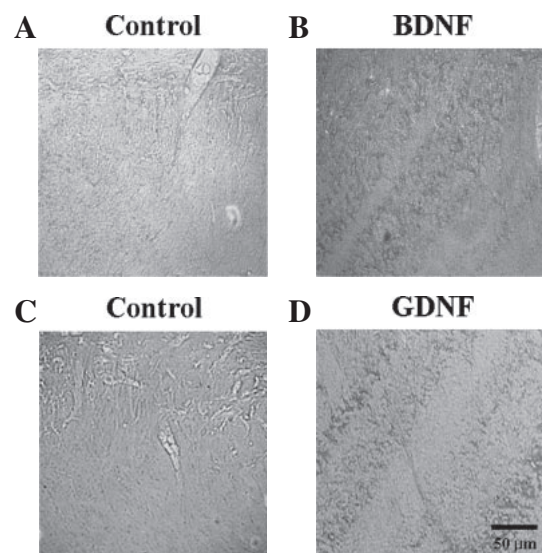


Fig. 5. Photomicrographs of BDNF (A, B) and GDNF (C, D) Immunostained Sections through the Substantia Nigra in Control and Detoxified RVS-Treated Animals.

Rats were treated with 10 mg/kg of RVS in saline for 4 h, as described in "Materials and Methods." Coronal sections (35 μ m) of the brain were cut on a cryostat and processed for immunohistochemistry. Photographs were captured with an inverted fluorescence microscope (Nikon, Eclipse TE 2000-U, Tokyo) at a magnification of 400 $\times$. Note the increased BDNF and GDNF immunoreactivity found in the animals treated with RVS. Scale bar = 50 μ m.

regulation of NTFs. The present study indicates for the first time that detoxified RVS extract can induce the expression of the BDNF and the GDNF gene both *in vitro* and *in vivo* at both the mRNA and the protein level. Many studies have suggested that flavonoids (such as fustin, fisetin, quercetin, butein sulfuretin) are the major active compounds responsible for the biological activity of RVS.^{22,29,30} Recently it was found that fustin, an important component of RVS, increases BDNF expression and attenuates β -amyloid-induced learning impairment in mice by modulating extracellular signal-

regulated kinase 1/2 (ERK1/2) and c-AMP response-element binding protein (CREB).³¹⁾ The flavonoid fisetin, another important component of RVS, also found in strawberries, has been reported to promote ERK-dependent long-term potentiation and to enhance memory in rat hippocampal slices by activating BDNF receptors.³²⁾ Emerging evidence suggests that dietary polyphenols such as resveratrol, catechins, sulforaphenes, and quinine hypericin have the remarkable ability to increase BDNF in the human brain.^{33,34)} Curcumin from the dietary supplement *Curcuma longa* has neuroprotective effects by activating BDNF in rodent cortical neurons.³⁵⁾ Carnosic acid, a bioactive polyphenol present in rosemary, has been found to induce BDNF in cultured dopaminergic cells.³⁶⁾ L-Theanine, a green-tea component, has been found to induce NTFs in SH-SY5Y cells.³⁷⁾ Xu *et al.*,³⁸⁾ have found that a dietary supplement, *Ginkgo biloba* extract EGB 761, increases BDNF in neuroblastoma cells. Royal jelly, a honey-bee secretion, stimulates GDNF mRNA expression in the hippocampus and cortex in the adult mouse. It has no effect on GDNF expression in the midbrain, but the effect on the striatum was not addressed.³⁹⁾ In this study, we found that RVS upregulated both NTFs, BDNF and GDNF, in the rat brain and in human dopaminergic cells, SH-SY5Y, in mRNA and protein levels in dose- and time-dependent manners. Moreover, our immunohistochemistry results indicate that RVS increases BDNF and GDNF expression in substantia nigra of the rat brain, suggesting that these effects are specific to RVS. Mounting evidence indicates that endogenous BDNF and GDNF expression can have profound clinical implications for the management of PD. These NTFs play key roles in organizational and activational processes by virtue of their capacity to regulate developmental processes, neuronal survival, morphology, and neuronal plasticity.^{40,41)} It has been suggested that alterations in NTF levels due to aging, genetic background, or other features contribute to neurodegeneration.²⁾ Pharmacological upregulation of endogenous NTFs can contribute to neuroprotection in neurodegenerative diseases such as PD. Therefore NTFs and the inducer of NTF biosynthesis would be expected to have enormous therapeutic potential for neuronal diseases.

It is well known that neurons in the brain are post-mitotic cells that cannot regenerate easily. They promptly undergo necrotic or apoptotic cell death once injured or damaged. Regeneration or sprouting of neurites depends on local guidance cues and target-derived trophic substances.⁴²⁾ Hence, the use of substances or factors that have trophic activity on neurons may be of benefits in regenerating damaged neurons. The present study indicates that oral administration to rats of RVS significantly increased mRNA levels of BDNF and GDNF and remarkably elevated the protein levels of these two neurotrophic factors. In this study, we used human dopaminergic cells, SH-SY5Y. This cell line is used extensively as a neuron model in many neurobiological, neurochemical, and neurotoxicological studies.^{43–48)} Our findings indicate that RVS is a potent inducer of BDNF and GDNF production in human dopaminergic cells SH-SY5Y, as indicated by enhanced expression of these NTFs. It has been reported that

various molecules regulate the expression of BDNF and GDNF in dopamine neurons. L-Theanine, a green tea component, transcription factor Pitx3, the antiexcitotoxic agent riluzole, the *N*-methyl-D-aspartate (NMDA)-receptor antagonist ifenprodil, and the mood stabilizers valproate, lithium, and anti-Parkinson monoamine oxidase (MAO)-B inhibitor rasagiline, have been found to increase BDNF and GDNF expression in cultured cells and tissues and to support the survival and differentiation of dopamine neurons.^{10,37,49–51)}

Although two NTFs, BDNF and GDNF, were upregulated by RVS, the extent of increase differed between them. The degrees of induction were significantly different at the transcriptional as well as the translational level, but temporal changes were similar among these NTFs at the two levels. In addition, there was clearly a quantitative difference between the BDNF and the GDNF mRNA and protein level following RVS treatment. This disparity is not surprising, since it is well known that the volume of mRNA resulting from gene expression commonly does not lead to equivalent quantities of the target protein.⁵²⁾ Furthermore, these discrepancies may have been due to differences in time course, mode of analysis, and most importantly, the experimental conditions.

In the present study, we found that RVS stimulated the secretion of BDNF and GDNF. It remains to be elucidated whether the present level of expression of these NTFs is sufficient to provide effective neuroprotection. However, many studies have found that similar levels of induction of these NTFs can offer neuroprotection in several models of neuronal injury. For example, oral administration of rasagiline (a promising drug for the treatment of PD) to 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP)-lesioned mice resulted in an elevation of NTF signaling and attenuated the loss of dopaminergic neurons from substantia nigra.⁵³⁾ Oral treatment with dietary supplement NT-020 (a proprietary formulation of blueberry, green tea, vitamin D3, and carnosine) upregulated the neurotrophic factors and protected against ischemic stroke in rats,⁵⁴⁾ but these treatments were administered over long durations. Pitx3-transfected astrocytes, carnosic acid, L-theanine, and triphenylmethyl chloride stimulated the secretion of NTFs and attenuated the oxidative damage induced by neurotoxins in the dopaminergic cells.^{36,37,50,55)} Earlier, we also found that the present level of induction of these NTFs can afford neuroprotection in some models of neuronal injury (data not shown). Taken together, these observations suggest that the degree of induction of NTF expression found in this study is likely to play a role in neuroprotective mechanisms. Additionally, the induction of NTFs observed for RVS is comparable to the induction of BDNF mRNA and protein levels by a neuroprotective agent, memantine, in the rat brain,⁵⁶⁾ and to the induction of GDNF in SH-SY5Y cells by rasagiline.⁴⁹⁾ Furthermore, oral administration of royal jelly for 3 d induced a similar level of GDNF in the mouse brain.³⁹⁾ In the present study, BDNF and GDNF immunoreactive neurons were identified in the substantia nigra of the rat brain, in agreement with the findings of previous studies.^{10,57–59)} The expression of NTFs due to memantine and rasagiline was transient. Transient expression

of BDNF and GDNF was also reported for pigment epithelium-derived factor (PEDF).⁶⁰⁾ In agreement with these studies, we observed transient increments of BDNF and GDNF. At the moment, it is not clear what role the transient expression of these NTFs might play, but it has been reported that transient delivery of neurotrophic factors is sufficient to sustain regenerated axons for prolonged periods within spinal-cord lesion sites.⁶¹⁾ This indicates that transient expression of NTFs by RVS may be effective for neuroprotection.

This study examined transcriptional and translational changes of the BDNF and the GDNF gene in response to treatment with RVS. The detailed molecular mechanism by which RVS treatment upregulates the expression of BDNF and GDNF is unknown, but studies have shown that activation of the neuronal survival pathway increases NTFs.⁶²⁾ Emerging evidence suggests that polyphenolic compounds promote neuronal survival.³⁴⁾ RVS contains many polyphenolic compounds. It is possible that these effects are initiated by RVS-induced release of NTFs from the rat brain and SH-SY5Y cells. Another possible mechanism by which RVS administration induces BDNF and GDNF in the rat brain is indirectly through the dopaminergic system. It has been reported that dopaminergic stimulation can upregulate the *in vivo* expression of BDNF and GDNF.^{10,63)} This is supported by our finding that treatment with RVS increases the expression of tyrosine hydroxylase (TH) in the rat brain.²⁵⁾ Moreover, previous studies indicate that the cAMP-response element (CRE) is a key element in the transcription of BDNF and GDNF.^{10,64)} RVS treatment has been found to increase the expression of target genes regulated by CREB, such as TH. Therefore, it is possible that RVS treatment increases the transcriptional activity of CRE, and consequently upregulates BDNF and GDNF expression.

Many studies have found that BDNF and GDNF exert distinct neurotrophic and pharmacological effects on both intact and injured nigrostriatal dopamine neurons *in vitro* and *in vivo*, but their inability to cross the blood-brain barrier (BBB) limits their clinical use. Therefore, a natural substance that can stimulate brain secretion of these NTFs might be useful in the development of therapeutic strategies for PD. RVS contains polyphenolic compounds and some polysaccharides. It has been reported that some polyphenols can penetrate the BBB.⁶⁵⁾ Moreover, it has been documented that some polysaccharides are indeed able to pass through the BBB and protect the brain from various pathological factors.^{66–68)} Since heparin has a polysaccharide backbone and has the potential to pass through the BBB to elicit its protective effects, there is a possibility that other neuroprotective polysaccharides also process similar properties. Currently it is unknown whether RVS can pass through the BBB. Further investigation is needed to determine this.

In conclusion, the present study indicates that detoxified RVS increases the expression of BDNF and GDNF in SH-SY5Y cells and the rat brain at both the mRNA and the protein level. Although the mechanisms by which detoxified RVS exerts its neuroprotective effect remain to be elucidated, our results indicate that RVS is a candidate for the treatment of neurodegenerative disorders such as PD.

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